

```
In [1]: import pandas as pd

In [2]: ssim_df = pd.read_csv('ssim_table_with_styles.csv')
        ssim_df

Out[2]:
   category content  exp1  exp2  exp3  exp4  exp5 name1 name2 name3 name4 name5
0  landscape  beaulieu  0.209986  0.054180  0.022443  0.153979  0.169131  K  K  K  K  K
1  landscape  beaulieu  0.214029  0.080033  0.025743  0.157171  0.334289  V  V  V  V  M
2  landscape  beaulieu  0.303134  0.172214  0.028252  0.266713  0.335573  P  P  P  P  P
3  still-life  skull  0.101914  0.051823  0.015587  0.091950  0.429772  K  K  K  K  P
4  still-life  skull  0.305680  0.189936  0.020415  0.297561  0.344851  M  M  M  M  M
5  still-life  skull  0.316816  0.195207  0.023217  0.297387  0.343552  T  T  T  T  T
6  portrait  girl  0.415137  0.260297  0.034836  0.411860  0.462338  P  P  P  P  P
7  portrait  girl  0.295497  0.136942  0.031002  0.328436  0.339268  T  T  T  T  T
8  portrait  girl  0.230094  0.102864  0.029521  0.216930  0.219434  V  V  V  V  V
9  portrait  man  0.235351  0.102899  0.027693  0.179743  0.241078  V  V  V  V  V
10 portrait  man  0.355389  0.156908  0.032182  0.315319  0.411497  T  T  T  T  T
11 portrait  man  0.465034  0.275813  0.033390  0.392104  0.449830  P  P  P  P  P
12 landscape  sea  0.477864  0.337286  0.036081  0.557083  0.541419  P  P  P  P  P
13 landscape  sea  0.188139  0.116533  0.028481  0.213139  0.504453  V  V  V  V  M
14 landscape  sea  0.091457  0.088491  0.024113  0.093289  0.094173  K  K  K  K  K
15 landscape  mountain  0.123257  0.062465  0.020705  0.118160  0.126175  K  K  K  K  K
16 landscape  mountain  0.386705  0.228977  0.028673  0.396105  0.453346  P  P  P  P  P
17 landscape  mountain  0.181916  0.094070  0.023798  0.155231  0.411111  V  V  V  V  M
18 still-life  still  0.414461  0.317972  0.036049  0.425128  0.448248  M  M  M  M  M
19 still-life  still  0.145544  0.080609  0.023099  0.139929  0.498059  K  K  K  K  P
20 still-life  still  0.358233  0.175674  0.030938  0.397688  0.427335  T  T  T  T  T
21 portrait  repin  0.264787  0.096421  0.025996  0.243571  0.281245  V  V  V  V  V
22 portrait  repin  0.402834  0.109913  0.029996  0.401873  0.445047  T  T  T  T  T
23 portrait  repin  0.478213  0.224162  0.030181  0.448358  0.508120  P  P  P  P  P
24 still-life  fish  0.216354  0.073500  0.023937  0.193486  0.230553  T  T  T  T  T
25 still-life  fish  0.211661  0.145739  0.028447  0.170802  0.220414  M  M  M  M  M
26 still-life  fish  0.159535  0.055543  0.020580  0.133579  0.247276  K  K  K  K  P

In [3]: import pandas as pd

parts = []

for i in range(1, 6):
    tmp = ssim_df[["category", "content", f"name{i}", f"exp{i}"]].copy()

    tmp = tmp.rename(columns={
        f"name{i}": "style",
        f"exp{i}": "value"
    })

    tmp["experiment"] = f"exp{i}"
    parts.append(tmp)

df_long = pd.concat(parts, ignore_index=True)
df_long

Out[3]:
   category content style value experiment
0  landscape  beaulieu  K  0.209986  exp1
1  landscape  beaulieu  V  0.214029  exp1
2  landscape  beaulieu  P  0.303134  exp1
3  still-life  skull  K  0.101914  exp1
4  still-life  skull  M  0.305680  exp1
...
130 portrait  repin  T  0.445047  exp5
131 portrait  repin  P  0.508120  exp5
132 still-life  fish  T  0.230553  exp5
133 still-life  fish  M  0.220414  exp5
134 still-life  fish  P  0.247276  exp5

135 rows x 5 columns

In [4]: df_long["normalized_value"] = (df_long["value"] / df_long.groupby("experiment")["value"].transform("mean") - 1)
df_long

Out[4]:
   category content style value experiment normalized_value
0  landscape  beaulieu  K  0.209986  exp1 -0.248960
1  landscape  beaulieu  V  0.214029  exp1 -0.234500
2  landscape  beaulieu  P  0.303134  exp1 0.084196
3  still-life  skull  K  0.101914  exp1 -0.635492
4  still-life  skull  M  0.305680  exp1 0.093303
...
130 portrait  repin  T  0.445047  exp5 0.262534
131 portrait  repin  P  0.508120  exp5 0.441463
132 still-life  fish  T  0.230553  exp5 -0.345955
133 still-life  fish  M  0.220414  exp5 -0.374719
134 still-life  fish  P  0.247276  exp5 -0.298515

135 rows x 6 columns

In [5]: df_long.groupby('category')['normalized_value'].describe()

Out[5]:
   count mean std min 25% 50% 75% max
category
landscape 45.0 -0.083616 0.463978 -0.732846 -0.417608 -0.175944 0.166262 1.284404
portrait 45.0 0.151298 0.333804 -0.377497 -0.086174 0.138285 0.440785 0.868058
still-life 45.0 -0.067682 0.392844 -0.655024 -0.374719 -0.121102 0.212286 1.153592

In [6]: import statsmodels.formula.api as smf

In [7]: model_no_category = smf.mixedlm(
    "normalized_value ~ C(experiment) + C(style)",
    data=df_long,
    groups=df_long["content"]
).fit(reml=False)
model_no_category.summary()

Out[7]:
Model: MixedLM Dependent Variable: normalized_value
No. Observations: 135 Method: ML
No. Groups: 9 Scale: 0.0453
Min. group size: 15 Log-Likelihood: 8.2548
Max. group size: 15 Converged: Yes
Mean group size: 15.0

Coef. Std.Err. z P>|z| [0.025 0.975]
Intercept -0.428 0.073 -5.859 0.000 -0.572 -0.285
C(experiment)[T.exp2] -0.000 0.058 -0.000 1.000 -0.114 0.114
C(experiment)[T.exp3] -0.000 0.058 -0.000 1.000 -0.114 0.114
C(experiment)[T.exp4] -0.000 0.058 -0.000 1.000 -0.114 0.114
C(experiment)[T.exp5] -0.150 0.059 -2.545 0.011 -0.265 -0.034
C(style)[T.M] 0.679 0.068 9.916 0.000 0.545 0.813
C(style)[T.P] 0.859 0.060 14.440 0.000 0.743 0.976
C(style)[T.T] 0.537 0.062 8.718 0.000 0.417 0.658
C(style)[T.V] 0.192 0.064 3.016 0.003 0.067 0.316
Group Var 0.020 0.053

In [8]: model_with_category = smf.mixedlm(
    "normalized_value ~ C(category) + C(experiment) + C(style)",
    data=df_long,
    groups=df_long["content"]
).fit(reml=False)
model_with_category.summary()

Out[8]:
Model: MixedLM Dependent Variable: normalized_value
No. Observations: 135 Method: ML
No. Groups: 9 Scale: 0.0453
Min. group size: 15 Log-Likelihood: 9.1547
Max. group size: 15 Converged: Yes
Mean group size: 15.0

Coef. Std.Err. z P>|z| [0.025 0.975]
Intercept -0.429 0.094 -4.570 0.000 -0.612 -0.245
C(category)[T.portrait] 0.092 0.114 0.807 0.420 -0.131 0.315
C(category)[T.still-life] -0.071 0.116 -0.612 0.541 -0.297 0.156
C(experiment)[T.exp2] -0.000 0.058 -0.000 1.000 -0.113 0.113
C(experiment)[T.exp3] -0.000 0.058 -0.000 1.000 -0.113 0.113
C(experiment)[T.exp4] -0.000 0.058 -0.000 1.000 -0.113 0.113
C(experiment)[T.exp5] -0.151 0.059 -2.564 0.010 -0.266 -0.035
C(style)[T.M] 0.685 0.069 9.931 0.000 0.550 0.820
C(style)[T.P] 0.847 0.060 14.123 0.000 0.730 0.965
C(style)[T.T] 0.531 0.062 8.501 0.000 0.408 0.653
C(style)[T.V] 0.176 0.064 2.744 0.006 0.050 0.302
Group Var 0.016 0.043

In [9]: from scipy.stats import chi2

lr = 2 * (model_with_category.llf - model_no_category.llf)
df_diff = (model_with_category.df_modelwc - model_no_category.df_modelwc)

p = chi2.sf(lr, df_diff)

print("p-value:", p)

p-value: 0.406612889999304

In [10]: import statsmodels.formula.api as smf

model_base = smf.mixedlm("normalized_value ~ C(category) + C(experiment) + C(style)",
    data=df_long,
    groups=df_long["content"])

result_base = model_base.fit(reml=False)

print(result_base.summary())

Mixed Linear Model Regression Results
=====
Model: MixedLM Dependent Variable: normalized_value
No. Observations: 135 Method: ML
No. Groups: 9 Scale: 0.0453
Min. group size: 15 Log-Likelihood: 9.1547
Max. group size: 15 Converged: Yes
Mean group size: 15.0

Coef. Std.Err. z P>|z| [0.025 0.975]
Intercept -0.429 0.094 -4.570 0.000 -0.612 -0.245
C(category)[T.portrait] 0.092 0.114 0.807 0.420 -0.131 0.315
C(category)[T.still-life] -0.071 0.116 -0.612 0.541 -0.297 0.156
C(experiment)[T.exp2] -0.000 0.058 -0.000 1.000 -0.113 0.113
C(experiment)[T.exp3] -0.000 0.058 -0.000 1.000 -0.113 0.113
C(experiment)[T.exp4] -0.000 0.058 -0.000 1.000 -0.113 0.113
C(experiment)[T.exp5] -0.151 0.059 -2.564 0.010 -0.266 -0.035
C(style)[T.M] 0.685 0.069 9.931 0.000 0.550 0.820
C(style)[T.P] 0.847 0.060 14.123 0.000 0.730 0.965
C(style)[T.T] 0.531 0.062 8.501 0.000 0.408 0.653
C(style)[T.V] 0.176 0.064 2.744 0.006 0.050 0.302
Group Var 0.016 0.043

In [11]: model_interaction = smf.mixedlm(
    "normalized_value ~ C(category) + C(experiment) + C(style)",
    data=df_long,
    groups=df_long["content"])

result_interaction = model_interaction.fit(reml=False)

print(result_interaction.summary())

Mixed Linear Model Regression Results
=====
Model: MixedLM Dependent Variable: normalized_value
No. Observations: 135 Method: ML
No. Groups: 9 Scale: 0.0431
Min. group size: 15 Log-Likelihood: 12.1621
Max. group size: 15 Converged: Yes
Mean group size: 15.0

Coef. Std.Err. z P>|z| [0.025 0.975]
Intercept -0.481 0.107 -4.488 0.000 -0.691 -0.271
C(category)[T.portrait] 0.205 0.144 1.431 0.152 -0.076 0.487
C(category)[T.still-life] -0.040 0.146 -0.274 0.784 -0.327 0.247
C(experiment)[T.exp2] 0.064 0.098 0.652 0.514 -0.128 0.256
C(experiment)[T.exp3] 0.107 0.098 1.095 0.274 -0.085 0.299
C(experiment)[T.exp4] 0.015 0.098 0.153 0.878 -0.177 0.207
C(experiment)[T.exp5] -0.097 0.101 -0.958 0.338 -0.296 0.102
C(style)[T.M] 0.687 0.068 10.093 0.000 0.554 0.821
C(style)[T.P] 0.856 0.060 14.206 0.000 0.737 0.974
C(style)[T.T] 0.535 0.061 8.722 0.000 0.415 0.656
C(style)[T.V] 0.182 0.065 2.827 0.005 0.056 0.309
C(category)[T.portrait]:C(experiment)[T.exp2] -0.209 0.138 -1.511 0.131 -0.481 0.062
C(category)[T.still-life]:C(experiment)[T.exp2] 0.018 0.138 0.127 0.899 -0.254 0.289
C(category)[T.portrait]:C(experiment)[T.exp3] -0.235 0.138 -1.696 0.090 -0.506 0.037
C(category)[T.still-life]:C(experiment)[T.exp3] -0.087 0.138 -0.626 0.531 -0.358 0.185
C(category)[T.portrait]:C(experiment)[T.exp4] -0.039 0.138 -0.281 0.779 -0.310 0.232
C(category)[T.still-life]:C(experiment)[T.exp4] -0.006 0.138 -0.044 0.965 -0.277 0.265
C(category)[T.portrait]:C(experiment)[T.exp5] -0.093 0.141 -0.661 0.508 -0.370 0.183
C(category)[T.still-life]:C(experiment)[T.exp5] -0.069 0.143 -0.481 0.631 -0.349 0.212
Group Var 0.016 0.044

In [12]: from scipy.stats import chi2

lr = 2 * (result_interaction.llf - result_base.llf)
df_diff = (result_interaction.df_modelwc - result_base.df_modelwc)
p_value = chi2.sf(lr, df_diff)

print(lr)
print(df_diff)
print("p-value:", p_value)

6.814870962455802
8
p-value: 0.645566035237427

In [13]: model_no_style = smf.mixedlm(
    "normalized_value ~ C(category) + C(experiment)",
    data=df_long,
    groups=df_long["content"]
)

result_no_style = model_no_style.fit(
    reml=False,
    method="lbfgs"
)

print(result_no_style.converged)

/opt/homebrew/anaconda3/lib/python3.12/site-packages/statsmodels/base/model.py:589: SingularMatrixWarning: Random effects covariance is singular
    return -self.score(params, *args) / nobs
True
/var/folders/_2/nrxg9rv974ddvc3x_62_jjlw000gn/T/ipykernel_29389/3156962138.py:7: ConvergenceWarning: The MLE may be on the boundary of the parameter space.
    result_no_style = model_no_style.fit()

In [14]: from scipy.stats import chi2

lr = 2 * (result_base.llf - result_no_style.llf)

df_diff = (result_base.df_modelwc - result_no_style.df_modelwc)

p = chi2.sf(lr, df_diff)

print(lr)
print(df_diff)
print("p-value:", p)

149.17129111359657
4
p-value: 3.064114797081872e-31

In [15]: print(result_no_style.summary())

Mixed Linear Model Regression Results
=====
Model: MixedLM Dependent Variable: normalized_value
No. Observations: 135 Method: ML
No. Groups: 9 Scale: 0.1479
Min. group size: 15 Log-Likelihood: -65.4309
Max. group size: 15 Converged: Yes
Mean group size: 15.0

Coef. Std.Err. z P>|z| [0.025 0.975]
Intercept -0.084 0.103 -0.811 0.418 -0.286 0.119
C(category)[T.portrait] 0.235 0.112 2.100 0.036 0.016 0.454
C(category)[T.still-life] 0.016 0.102 0.142 0.887 -0.283 0.235
C(experiment)[T.exp2] -0.000 0.105 -0.000 1.000 -0.205 0.205
C(experiment)[T.exp3] 0.000 0.105 0.000 1.000 -0.205 0.205
C(experiment)[T.exp4] -0.000 0.105 -0.000 1.000 -0.205 0.205
C(experiment)[T.exp5] -0.000 0.105 -0.000 1.000 -0.205 0.205
Group Var 0.009 0.024
```